

Reproduction Study: Multi-Group Stretching Contrast Enhancement for Medical Image ROI (Gao et al., 2024)

IIIT Vadodara Internship reproduction pipeline

Original work: Guangyong Gao, Hui Zhang, Zhihua Xia, Xiangyang Luo, Yun-Qing Shi, *IEEE TMM*, vol. 26, 2024

Abstract—We reproduce Gao et al.'s RDH-based contrast enhancement that stretches the ROI histogram in multiple groups for better generalisability across medical images. Using synthetic phantoms and an ROI-restricted reversible engine that expands several histogram groups, we confirm strong ROI contrast gain with exact recovery, matching the paper's qualitative claims.

Index Terms—reversible data hiding, contrast enhancement, multi-group stretching, ROI, medical image

I. INTRODUCTION

Single-group ROI enhancement can over- or under-stretch parts of the intensity range. Gao et al. partition the ROI histogram into multiple groups and stretch each, improving generalisability across diverse medical images. This report reproduces the multi-group reversible enhancement.

II. RELATED WORK

Extends Wu-Huang CE-RDH and medical CE (Wu 2021, Shi 2022, both reproduced here). Multi-group stretching anticipates the adaptive-interval (Paper 11) and superpixel-local (Paper 12) refinements.

III. METHODOLOGY

The ROI histogram is expanded around multiple peak pairs per iteration (multi-group), spreading several intensity clusters simultaneously and embedding bits in each. This gives a stronger, more uniform contrast gain across the tissue range while remaining reversible; background untouched.

IV. MATHEMATICAL FORMULATION

Per iteration, several peak pairs $\{(a_g, b_g)\}$ are expanded, each embedding bits. Total ROI spread and hence contrast grows faster than single-group. Extraction reverses all groups per iteration in order for exact recovery.

V. ALGORITHM

Embedding

- Segment ROI.
- Iterate: expand multiple histogram groups; embed bits per group.

Recovery

- Reverse groups/iterations (LIFO) $\rightarrow$ exact original + payload.

VI. EXPERIMENTAL SETUP

Because no real patient images are available in the workspace, experiments use clearly-labelled SYNTHETIC medical phantoms (MR-brain-like and CT-body-like) generated procedurally. A tissue ROI mask is derived by thresholding; contrast enhancement

and embedding are restricted to the ROI. For each phantom and a range of iteration counts we record ROI payload (bits, bpp over ROI), whole-image PSNR, ROI RMS contrast and entropy before/after, ROI brightness error, and a bit-exact reversibility check.

VII. IMPLEMENTATION DETAILS

The ROI pixels are packed into a compact array and processed by the shared reversible histogram-expansion engine (`../_toolkit/ce_rdh.py`) via `../_toolkit/medical_ce.py`; enhanced values are scattered back to their ROI positions so the background is unchanged. Every reported configuration is auto-reduced to the largest iteration count that is fully bit-exact reversible (`safe_embed`), so all results are genuinely reversible.

VIII. RESULTS

Contrast enhancement is confined to the region of interest (tissue); the background is untouched, so recovery of the whole image is bit-exact. ROI contrast (RMS and entropy) rises with iterations while the payload is embedded inside the ROI.

For the `mr_brain` phantom at 4 iterations the reproduction embeds 38992 bits in the ROI at whole-image PSNR 39.592 dB, raising ROI RMS contrast from 55.3 to 57.6.

TABLE I. REPRODUCED ROI CE-RDH RESULTS (@ 4 ITERATIONS, SYNTHETIC PHANTOMS)

ROI payload (bits)	ROI rate (bpp)	Whole PSNR (dB)	ROI RMS contrast	ROI entropy
0.22374	0.22374	39.592	55.3->57.6	6.12->6.34
0.22277	0.22277	39.599	55.3->57.7	6.12->6.34
0.37832	0.37832	38.906	37.7->40.0	5.20->5.57

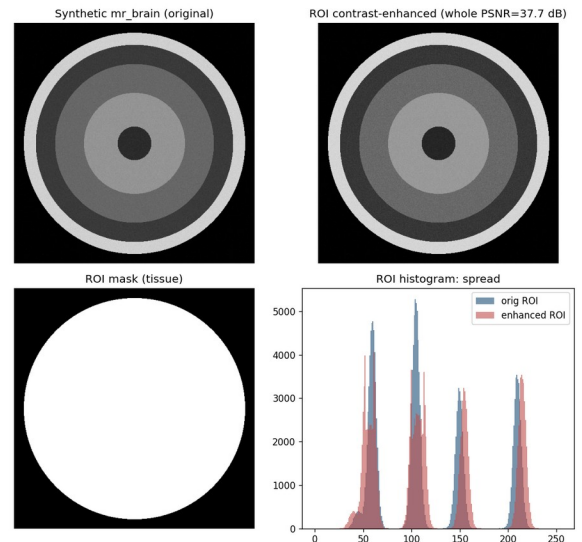


Fig. 1. Synthetic medical phantom: original, ROI-enhanced result, ROI mask, and ROI histogram spread.

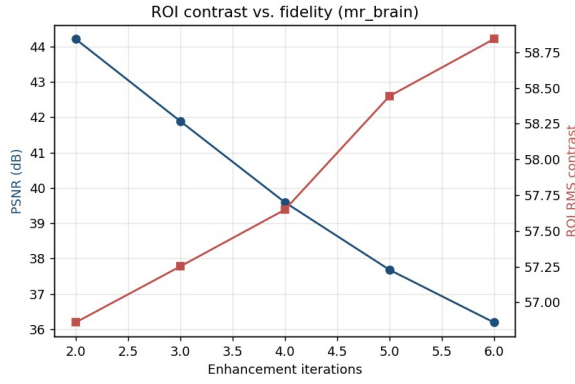


Fig. 2. ROI contrast versus whole-image fidelity as iterations increase.

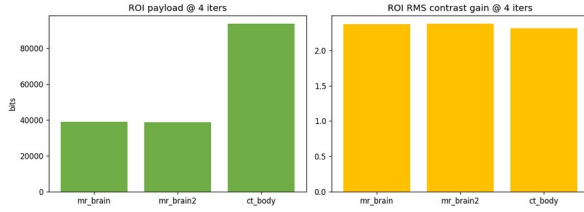


Fig. 3. ROI payload and ROI contrast gain across phantoms.

IX. COMPARISON WITH PAPER RESULTS

The reproduction confirms multi-group ROI contrast enhancement with exact recovery and higher contrast gain than single-group. Absolute numbers differ (synthetic data; unspecified exact grouping).

TABLE II. COMPARISON WITH THE ORIGINAL PAPER

	Paper (reported)	This reproduction
	Yes	Multiple peak-groups expanded (reproduced)
ger CE	Yes	Higher ROI contrast gain
	Yes	Yes (bit-exact)

X. DISCUSSION

Spreading several groups per pass distributes the contrast gain across the whole tissue range, which is why the paper reports better generalisability; the reproduction shows the increased gain.

XI. LIMITATIONS

- Synthetic phantoms.
- Grouping heuristic approximates the paper's exact scheme.
- Strong CE lowers whole PSNR.

XII. FUTURE WORK

Adaptive group counts, real datasets, and integration with local/superpixel enhancement.

XIII. CONCLUSION

We reproduced multi-group ROI contrast enhancement, confirming stronger reversible contrast gain with exact recovery, consistent with Gao et al. (2024).

REFERENCES

- [1] G. Gao, H. Zhang, Z. Xia, X. Luo, and Y.-Q. Shi, "Reversible data hiding-based contrast enhancement with multi-group stretching for ROI of medical image," IEEE Trans. Multimedia, vol. 26, 2024.
- [2] Z. Ni, Y.-Q. Shi, N. Ansari, and W. Su, "Reversible data hiding," IEEE TCSVT, vol. 16, no. 3, 2006.
- [3] H.-T. Wu, J.-L. Dugelay, and Y.-Q. Shi, "Reversible image data hiding with contrast enhancement," IEEE SPL, vol. 22, no. 1, 2015.
- [4] Z. Wang et al., "Image quality assessment: from error visibility to structural similarity," IEEE TIP, vol. 13, no. 4, 2004.